

Kaustubh Tripathi

Software Engineer

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SUMMARY

Software Engineer with 1 year of experience spanning front-end and backend development — building UI features in JavaScript at Tata Consultancy Services, alongside backend systems in Java and Spring Boot.

Skilled in JavaScript, TypeScript, React, Node.js, and Vite-based tooling from building full-stack projects, with strong REST API and component-driven UI experience.

Currently contributing to fintech applications at TCS across front-end feature work, backend services, and performance tuning.

SKILLS

Programming Languages: JavaScript, TypeScript, Java, SQL, C++

Frameworks & Libraries: React.js, Node.js, Express.js, Tailwind CSS, Spring Boot, Spring Security (JWT, OAuth2)

Databases: PostgreSQL, Redis (Caching), MongoDB, MySQL

Tools & Technologies: Vite, Git, Docker, Docker Compose, Postman, REST API, Webhooks, Microservices, Linux

WORK EXPERIENCE

Tata Consultancy Services Ltd.

Bangalore, Karnataka

Product Engineer

June 2025 - Present

- Contributed to full-stack development of a fintech/banking module — building front-end features in JavaScript and backend services in Java and Spring Boot — ensuring scalability and maintainability.
- Optimized database performance by identifying bottlenecks and applying SQL query tuning and indexing, improving execution efficiency of critical processes.
- Collaborated with cross-functional teams to implement new features and bug fixes, actively participating in code reviews and maintaining coding standards.
- Assisted in integrating new functionalities into legacy systems, ensuring minimal regression and high system stability.
- Took ownership of debugging production issues by analysing logs and tracing API failures, reducing issue resolution time

PROJECTS

Skill Barter Platform: [Demo](#) | [GitHub](#)

- Engineered a peer-to-peer skill exchange platform with a matching engine that classifies users as Perfect (mutual) or Partial (one-way) matches, sorted by geospatial proximity using MongoDB 2dsphere indexes
- Built a real-time chat system using Socket.io with room-based WebSocket architecture, supporting typing indicators, read receipts, and a two-way trade lifecycle requiring mutual consent for acceptance and completion
- Implemented dual-token JWT authentication (15-min access + 7-day refresh) with silent renewal via Axios interceptors, OTP-based email verification, and API rate limiting for brute-force prevention
- Deployed full-stack application on Render and Vercel with environment-driven configuration, CORS whitelisting, and bcrypt password hashing for production-grade security

LeetCode Agent Tracker: [GitHub](#)

- Built a full-stack LeetCode prep platform (React, Express, MongoDB) with problem tracking, a Leitner-style spaced-repetition scheduler, and an analytics dashboard covering topic heatmaps and company readiness
- Built an autonomous weekly-plan agent using Gemini function-calling — model independently selects from 4 database tools across up to 8 iterations, then outputs a rationale-backed study plan
- Built a Socratic hint tutor withholding solutions via graduated multi-turn hints, plus a post-mortem agent that reviews solved code to flag suboptimal time/space complexity
- Logged every agent invocation — tool calls, reasoning trace, latency — to a queryable collection for full auditability into agent decisions

EDUCATION

Bachelor of Technology in Computer Science and Engineering

Oct 2020- June 2024 | 7.9 CGPA

Madan Mohan Malaviya University of Technology, Gorakhpur, Uttar Pradesh